

CSE-251

ELECTRONIC CIRCUITS

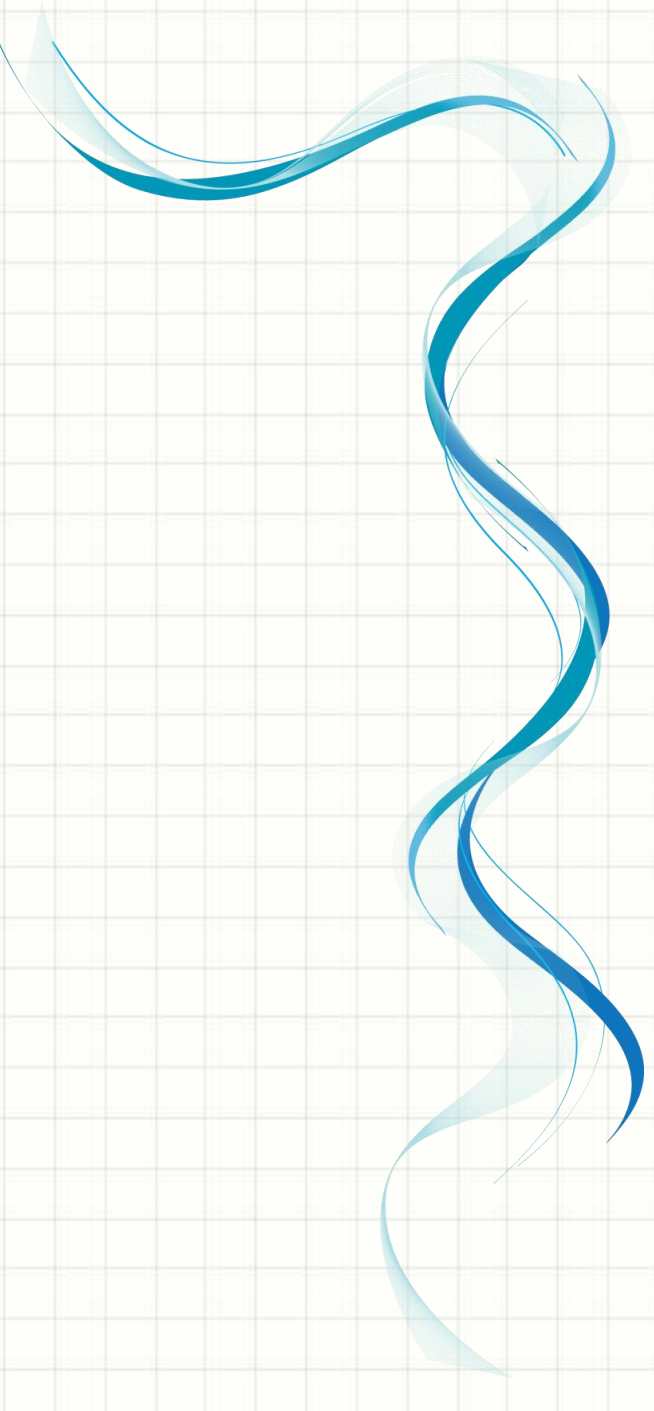
LECTURE-16

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Today's Overview:

- ❑ *Constant Gain Multipliers*
- ❑ *Multiple Stage Gains*

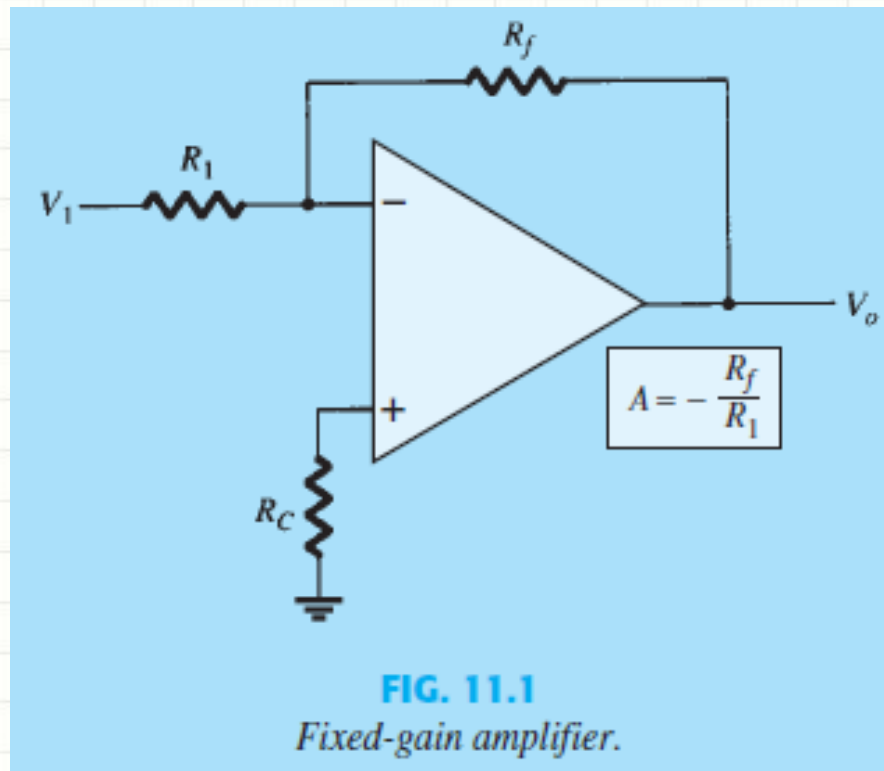


A decorative blue wavy line with a light blue shadow, flowing vertically on the left side of the slide.

Constant Gain Multipliers

Constant Gain/Fixed Gain Multipliers:

□ For inverting amplifier:



$$A = -\frac{R_f}{R_1}$$

Problem:

EXAMPLE 11.1 Determine the output voltage for the circuit of Fig. 11.2 with a sinusoidal input of 2.5 mV.

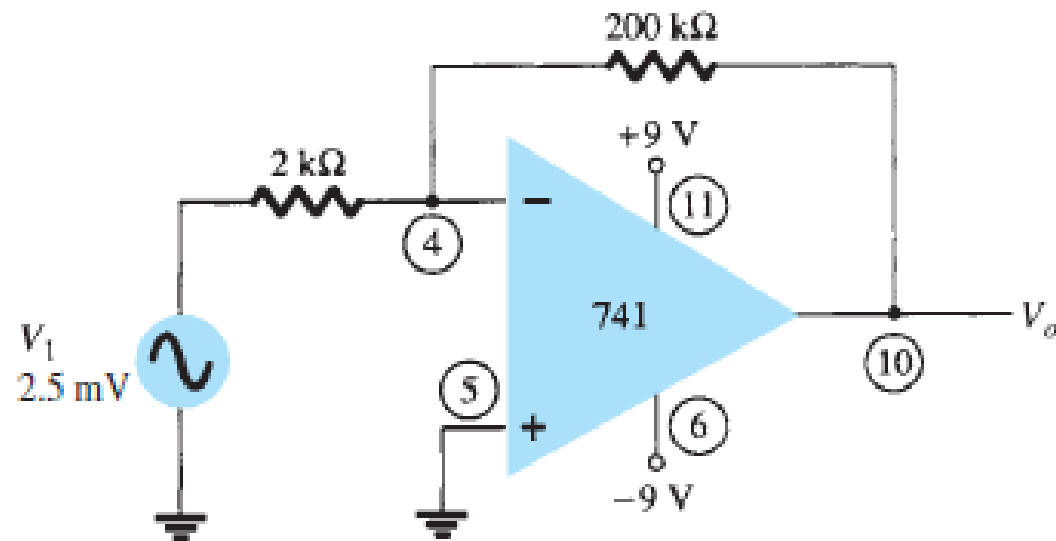


FIG. 11.2

Circuit for Example 11.1.

Problem:

Solution: The circuit of Fig. 11.2 uses a 741 op-amp to provide a constant or fixed gain, calculated from Eq. (11.1) to be

$$A = -\frac{R_f}{R_1} = -\frac{200 \text{ k}\Omega}{2 \text{ k}\Omega} = -100$$

The output voltage is then

$$V_o = AV_i = -100(2.5 \text{ mV}) = -250 \text{ mV} = -0.25 \text{ V}$$

Constant Gain/Fixed Gain Multipliers:

□ For non-inverting amplifier:

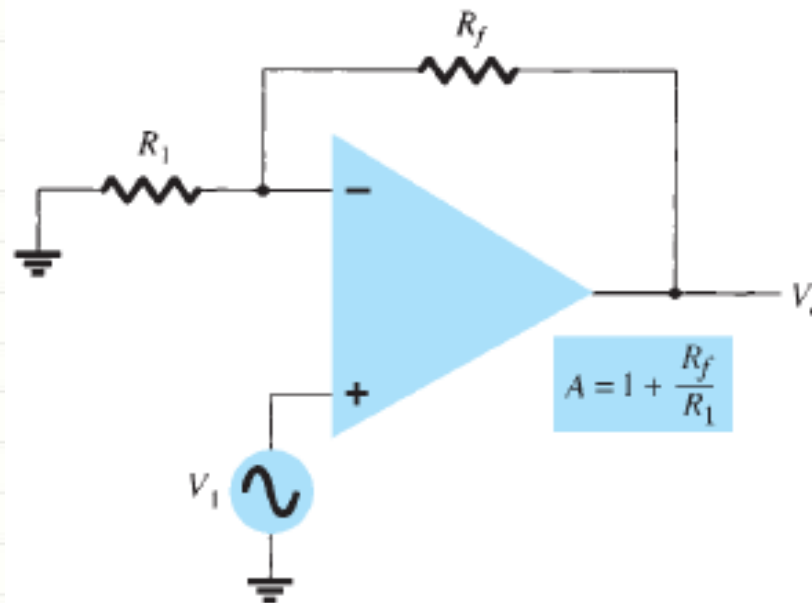


FIG. 11.3

Noninverting fixed-gain amplifier.

$$A = 1 + \frac{R_f}{R_1}$$

Problem:

EXAMPLE 11.2 Calculate the output voltage from the circuit of Fig. 11.4 for an input of $120 \mu\text{V}$.

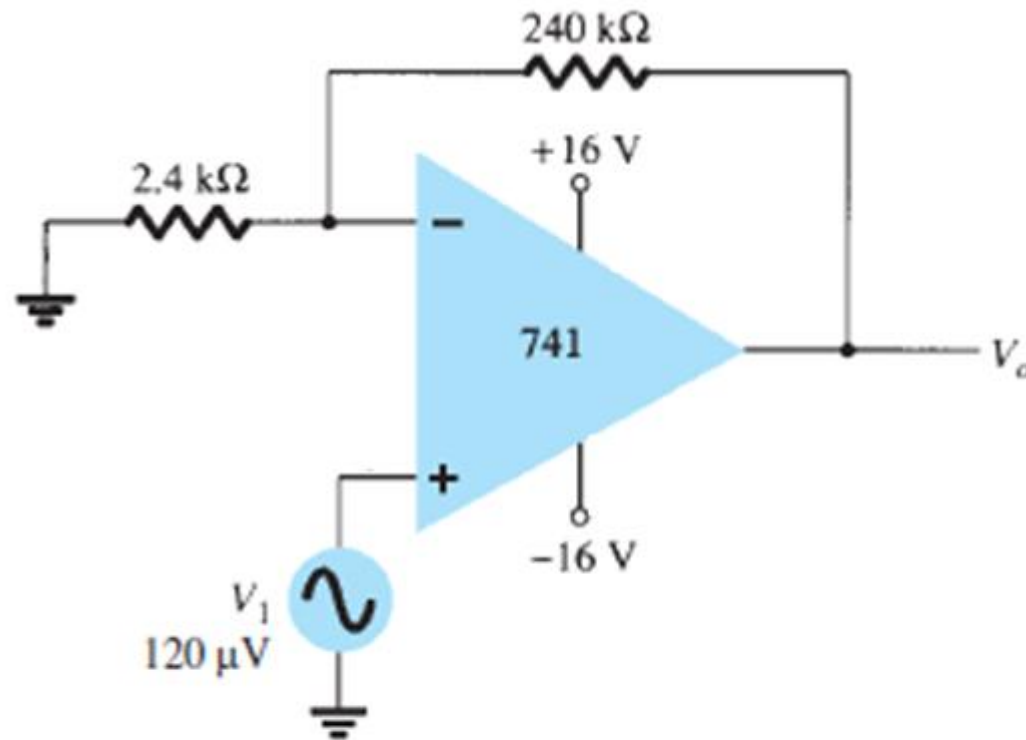


FIG. 11.4
Circuit for Example 11.2.

Problem:

Solution: The gain of the op-amp circuit is calculated using Eq. (11.2) to be

$$A = 1 + \frac{R_f}{R_1} = 1 + \frac{240 \text{ k}\Omega}{2.4 \text{ k}\Omega} = 1 + 100 = 101$$

The output voltage is then

$$V_o = AV_i = 101(120 \mu\text{V}) = \mathbf{12.12 \text{ mV}}$$

Multiple Stage Gains



Multiple Stage Gains:

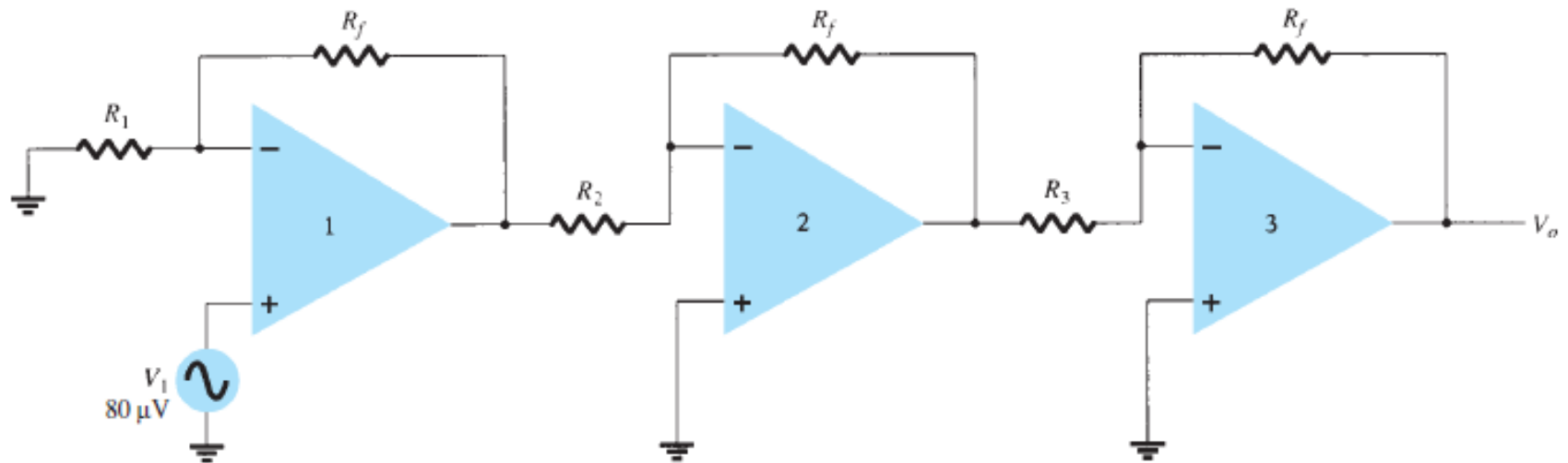


FIG. 11.5

Constant-gain connection with multiple stages.

$$A = A_1 A_2 A_3$$

where $A_1 = 1 + R_f/R_1$, $A_2 = -R_f/R_2$, and $A_3 = -R_f/R_3$.

Problem:

EXAMPLE 11.3 Calculate the output voltage using the circuit of Fig. 11.5 for resistor components of value $R_f = 470\text{ k}\Omega$, $R_1 = 4.3\text{ k}\Omega$, $R_2 = 33\text{ k}\Omega$, and $R_3 = 33\text{ k}\Omega$ for an input of $80\text{ }\mu\text{V}$.

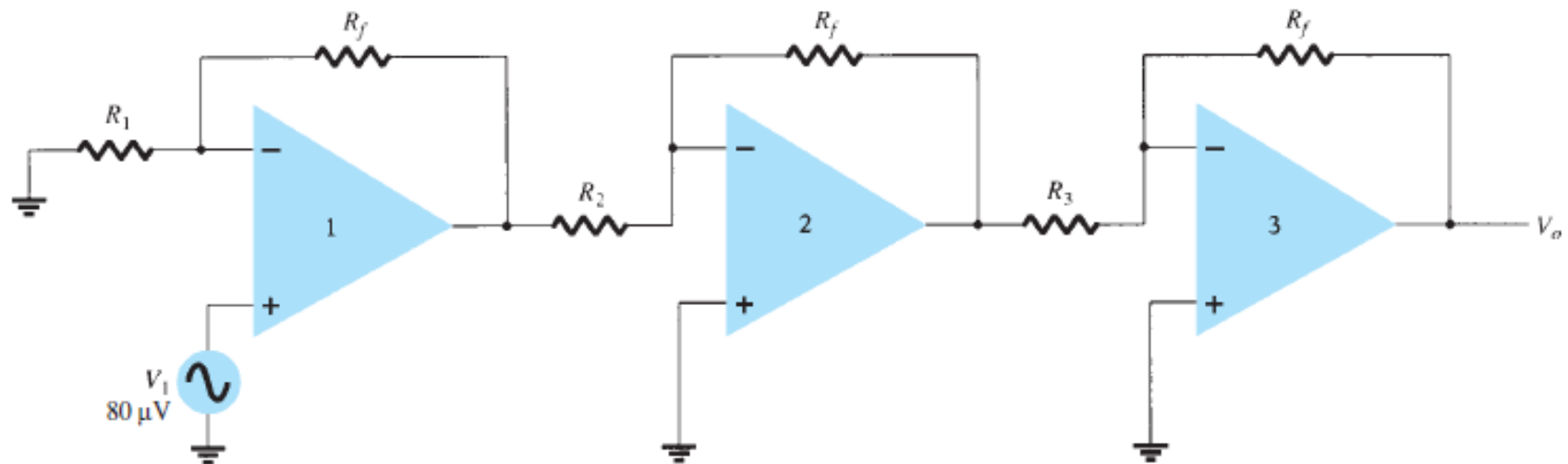


FIG. 11.5

Constant-gain connection with multiple stages.

Problem:

Solution: The amplifier gain is calculated to be

$$\begin{aligned} A &= A_1 A_2 A_3 = \left(1 + \frac{R_f}{R_1}\right) \left(-\frac{R_f}{R_2}\right) \left(-\frac{R_f}{R_3}\right) \\ &= \left(1 + \frac{470 \text{ k}\Omega}{4.3 \text{ k}\Omega}\right) \left(-\frac{470 \text{ k}\Omega}{33 \text{ k}\Omega}\right) \left(-\frac{470 \text{ k}\Omega}{33 \text{ k}\Omega}\right) \\ &= (110.3)(-14.2)(-14.2) = 22.2 \times 10^3 \end{aligned}$$

so that

$$V_o = AV_i = 22.2 \times 10^3 (80 \mu\text{V}) = 1.78 \text{ V}$$

Problem:

EXAMPLE 11.5 Show the connection of three op-amp stages using an LM348 IC to provide outputs that are 10, 20, and 50 times larger than the input. Use a feedback resistor of $R_f = 500 \text{ k}\Omega$ in all stages.

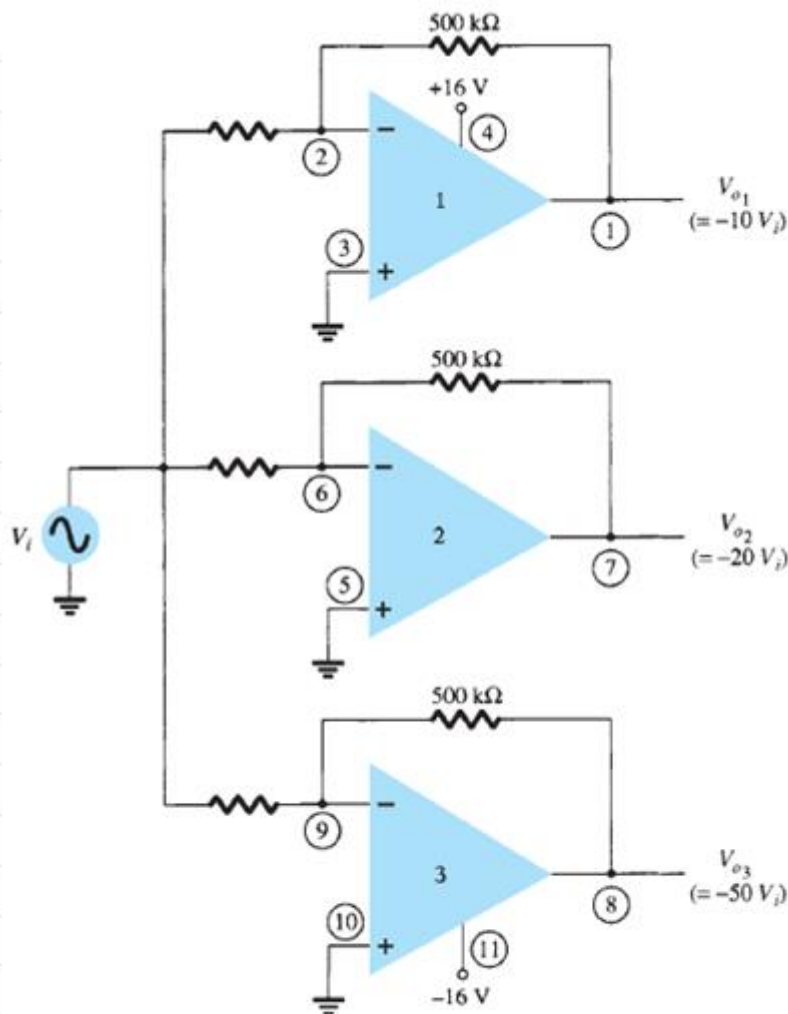


FIG. 11.7

Circuit for Example 11.5 (using LM348).

Problem:

Solution: The resistor component for each stage is calculated to be

$$R_1 = -\frac{R_f}{A_1} = -\frac{500 \text{ k}\Omega}{-10} = 50 \text{ k}\Omega$$

$$R_2 = -\frac{R_f}{A_2} = -\frac{500 \text{ k}\Omega}{-20} = 25 \text{ k}\Omega$$

$$R_3 = -\frac{R_f}{A_3} = -\frac{500 \text{ k}\Omega}{-50} = 10 \text{ k}\Omega$$

Problem:

EXAMPLE 11.4 Show the connection of an LM124 quad op-amp as a three-stage amplifier with gains of $+10$, -18 , and -27 . Use a $270\text{-k}\Omega$ feedback resistor for all three circuits. What output voltage will result for an input of $150\text{ }\mu\text{V}$?

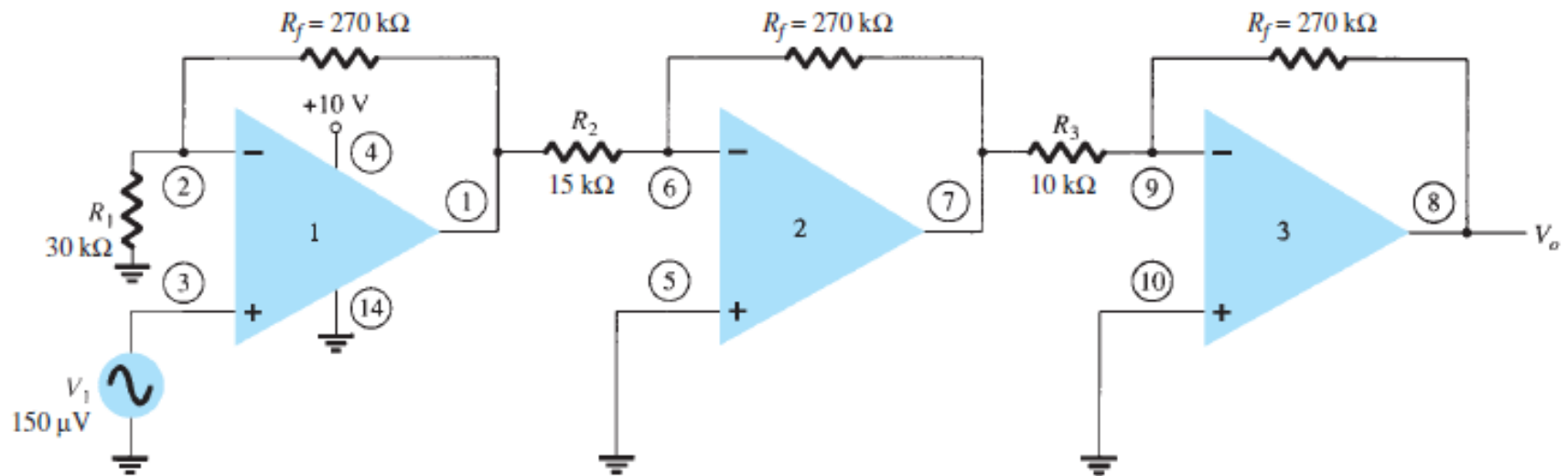


FIG. 11.6

Circuit for Example 11.4 (using LM124).

Problem:

Solution: For the gain of +10,

$$A_1 = 1 + \frac{R_f}{R_1} = +10$$

$$\frac{R_f}{R_1} = 10 - 1 = 9$$

$$R_1 = \frac{R_f}{9} = \frac{270 \text{ k}\Omega}{9} = 30 \text{ k}\Omega$$

For the gain of -18,

$$A_2 = -\frac{R_f}{R_2} = -18$$

$$R_2 = \frac{R_f}{18} = \frac{270 \text{ k}\Omega}{18} = 15 \text{ k}\Omega$$

For the gain of -27,

$$A_3 = -\frac{R_f}{R_3} = -27$$

$$R_3 = \frac{R_f}{27} = \frac{270 \text{ k}\Omega}{27} = 10 \text{ k}\Omega$$

The circuit showing the pin connections and all components used is given in Fig. 11.6. For an input of $V_1 = 150 \mu\text{V}$, the output voltage is

$$\begin{aligned} V_o &= A_1 A_2 A_3 V_1 = (10)(-18)(-27)(150 \mu\text{V}) = 4860(150 \mu\text{V}) \\ &= 0.729 \text{ V} \end{aligned}$$



QUESTIONS?

SEE YOU IN THE NEXT CLASS

